



Discrete Distributions



Introduction

There are three discrete probability distributions that serve as models for a large number of applications.

1. The Binomial Distribution
2. The Poisson Distribution
3. The Hypergeometric Distribution

What is Binomial Random Variables?



Example

Coin toss is a good example of Binomial random variable

Some other examples

1. A sociologist is interested in the proportion of elementary school teachers who are men.
2. A soft-drink marketer is interested in the proportion of cola drinkers who prefer her brand.
3. A geneticist is interested in the proportion of the population who possess a gene linked to Alzheimer's disease.



Definition

A binomial experiment is one that has these five characteristics:

1. The experiment consists of n identical trials.
2. Each trial results in one of two outcomes. For lack of a better name, the one outcome is called a success, **S**, and the other a failure, **F**.
3. The probability of success on a single trial is equal to p and remains the same from trial to trial. The probability of failure is equal to $(1 - p) = q$
4. The trials are independent.
5. We are interested in $\underline{x}$, the number of successes observed during the n trials, for $x = 0, 1, 2, \dots, n$.



Example

Suppose there are approximately **1,000,000** adults in a county and an unknown proportion p favor term limits for politicians. A sample of 1000 adults will be chosen in such a way that every one of the 1,000,000 adults has an equal chance of being selected, and each adult is asked whether he or she favors term limits. Is this a binomial experiment?

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1. A “trial” is the choice of a single adult from the 1,000,000 adults in the county. This sample consists of n 1000 identical trials.
2. Since each adult will either favor or not favor term limits, there are two outcomes that represent the “successes” and “failures” in the binomial experiment.
3. The probability of success, p , is the probability that an adult favors term limits. Does this probability remain the same for each adult in the sample?
4. The independence of the trials is guaranteed because of the large group of adults from which the sample is chosen. The probability of an adult favoring term limits does not change depending on the responses of previously chosen people.
5. The random variable x is the number of adults in the sample who favor term limits.



Another Example

A patient fills a prescription for a **10-day** regimen of **2 pills daily**. Unknown to the pharmacist and the patient, the 20 tablets consist of **18 pills of the prescribed medication** and **2 pills that are the generic equivalent of the prescribed medication**. The patient selects two pills at random for the first day's dosage. If we check the selection and record the number of pills that are generic, is this a binomial experiment?



Rule of Thumb

- If the sample size is large relative to the population size—in particular, if $n/N \geq .05$, then the resulting experiment is not binomial.



Example

Find $P(x = 2)$ for a binomial random variable with $n=10$ and $p=.1$.



Example

Over a long period of time, it has been observed that a professional basketball player can make a free throw on a given trial with probability equal to **.8**. Suppose he shoots four free throws.

1. What is the probability that he will make exactly two free throws?
2. What is the probability that he will make at least one free throw?

Cumulative Binomial Table

$$n = 5$$
[illegible]